

The dark-photon production cross section at a reactor

Lecture 1: one exact formula, three notes, eight papers, and which differences are errors

E. L. De Paoli

written with agent assistance (Claude Opus 5)

21 September 2026

Abstract

The documents in `apuntes/` do not compute the same thing. The Spanish note (`Dark_Photon_cross_section-1.pdf`) derives the tree-level cross section for $\gamma e^- \rightarrow A' e^-$ and gets the massive case wrong: the squared amplitude, the kinematics and the kinematic range contain errors, and its final $d\sigma/dE_{A'}$ is off by up to a factor 2 pointwise and crosses a propagator pole inside its own integration range for $m_{A'} = 1$ MeV. Each error has a one-line fix (Section 6). The *v2* report (`report_park_production_v2`) has the tree-level cross section exactly right: I re-derived it with explicit Dirac matrices and it agrees with the *v2* formula, with deNiverville–Lee–Lee and with Smirnov et al. to 2×10^{-13} . Its fold with the reactor spectrum, however, is not what Park’s source equation gives: both *v2*’s Figure 2 and Park’s Figure 1 have a tail equal to the photon spectrum itself, while the Compton-like convolution is 4–8 times lower above 2 MeV. The *v3* report (`report_darkphoton_v3`) uses a different mechanism (in-medium oscillation, Danilov et al.); it is not another way of doing the same calculation, it is the dominant production channel, and Park’s formula omits it. In TEXONO’s 3–8 MeV window the two mechanisms differ by a factor 7.7 at the same ϵ (Figure 3). Park’s 2/3 detection factor is the unpolarized average and is wrong for the beam a reactor produces; the correct factor is ≈ 1 for $m_{A'} \lesssim 0.1$ MeV and between 0.8 and 1 up to 1 MeV.

1 The question

Three things are asked. (i) Do the documents in `apuntes/` differ in the theoretical approach to the cross section? (ii) Do the *final* formulas needed for the number of dark photons at a detector differ? (iii) Are the differences errors? The answers are: yes; yes; and mostly yes, with one exception that is not an error but an omission of physics.

Everything numerical below was produced by `work/check_cross_section.py` (every block ends in an `assert`) and `work/figures_cross_section.py` (marimo). The run receipt is in Section 11. What I derived and what I recalled is stated at each step.

2 Setup

Process, notation and interaction (recalled, standard):

$$\gamma(k) + e^-(p) \rightarrow A'(q) + e^-(p'), \quad m \equiv m_e, \quad M \equiv m_{A'}, \quad \mathcal{L}_{\text{int}} = -e \bar{\psi} \gamma^\mu \psi (A_\mu + \epsilon A'_\mu), \quad e^2 = 4\pi\alpha. \quad (1)$$

The electron is free and at rest. Two tree diagrams (*s*- and *u*-channel electron exchange). Invariants:

$$s = (p+k)^2 = m^2 + 2mE, \quad u = (p-q)^2 = m^2 + M^2 - 2mx, \quad t = (p-p')^2 = 2m(x-E), \quad (2)$$

with E the photon energy and x the A' energy in the lab. I use the propagator denominators

$$a \equiv s - m^2 = 2mE, \quad b \equiv u - m^2 = M^2 - 2mx, \quad c \equiv m^2, \quad d \equiv M^2. \quad (3)$$

3 Kinematics (derived, exact)

From $(p + k - q)^2 = m^2$ with $Q = |\vec{q}| = \sqrt{x^2 - M^2}$:

$$M^2 + 2m(E - x) - 2E(x - Q \cos \theta) = 0 \implies \cos \theta = \frac{(x - m)E + mx - M^2/2}{EQ}. \quad (4)$$

This is transcendental in x at fixed θ because $Q \neq x$. The endpoints of x come from the CM two-body momentum boosted to the lab:

$$x_{\pm} = \frac{(E + m)(s + M^2 - m^2) \pm E\sqrt{\lambda(s, m^2, M^2)}}{2s}, \quad \lambda(s, m^2, M^2) = [s - (m + M)^2][s - (m - M)^2], \quad (5)$$

and the threshold is $s \geq (m + M)^2$:

$$E_{\text{th}} = M + \frac{M^2}{2m} : \quad 0.1098, 0.7446, 1.9785 \text{ MeV for } M = 0.1, 0.5, 1 \text{ MeV}. \quad (6)$$

The threshold is *not* $E = M$. At $M = 1$ MeV it is twice that.

4 The squared amplitude (derived twice)

With $\Gamma^{\nu\mu} = \gamma^\nu(\not{p}_s + m)\gamma^\mu/a + \gamma^\mu(\not{p}_u + m)\gamma^\nu/b$, $r_s = p + k$, $r_u = p - q$, the amplitude is $\mathcal{M} = \epsilon e^2 \bar{u}(p')\Gamma^{\nu\mu}u(p)\varepsilon_\mu(k)\xi_\nu^*(q)$. Averaging over the two photon polarizations and two electron spins, summing over final spins and the three A' polarizations:

$$|\overline{\mathcal{M}}|^2 = \epsilon^2 e^4 F, \quad \boxed{F = -2\left(\frac{a}{b} + \frac{b}{a}\right) + 4(2c + d)\left[\left(\frac{1}{a} + \frac{1}{b}\right) + c\left(\frac{1}{a} + \frac{1}{b}\right)^2 - \frac{d}{ab}\right]} \quad (7)$$

This is eq. (36) of the *v2* report. I did not take it on trust. My check builds the 4×4 Dirac matrices, the explicit polarization vectors (two for the photon, $\varepsilon_{1,2}$; three for the A' , including $\xi_L = (Q, x\hat{q})/M$), forms Γ numerically, and evaluates $\frac{1}{4} \sum \text{Tr}[(\not{p}' + m)\Gamma(\not{p} + m)\bar{\Gamma}]$ at 60 random on-shell lab points with $M \in [0.05, 1.5]$ MeV, $E \leq 10$ MeV. No trace identities are used. Results:

- explicit polarization sum = covariant $(-g_{\mu\mu'})(-g_{\nu\nu'})$ sum to 2×10^{-13} (the Ward identity $q_\nu \mathcal{M}^\nu = 0$ holds, so the $q_\nu q_{\nu'}/M^2$ term drops);
- numeric = eq. (7) to 2×10^{-13} ;
- numeric = deNiverville–Lee–Lee eq. (18)–(20) to 2×10^{-13} ;
- $d\sigma/dx$ from eq. (7) = Smirnov et al. eq. (3) (with $Z = 1$, $g_X = \epsilon e$, $x_S = 1 - x/E + M^2/2mE$) to 9×10^{-16} .

Three independent published forms and one explicit-matrix computation agree. Eq. (7) is the tree-level cross section. At $d = 0$ it is Klein–Nishina, $F = 2[x/E + E/x - \sin^2 \theta]$ (checked with sympy).

5 Cross sections

Recalled: $d\sigma/dt = |\overline{\mathcal{M}}|^2/[16\pi(s - m^2)^2]$. Derived: $dt/dx = 2m$. Hence

$$\boxed{\frac{d\sigma}{dx} = \frac{\epsilon^2 e^4}{32\pi m E^2} F(a, b, c, d), \quad a = 2mE, \quad b = M^2 - 2mx, \quad x \in [x_-, x_+]} \quad (8)$$

Total: $\sigma = \int_{x_-}^{x_+} dx d\sigma/dx$; the *v2* closed form $G(b)$ (its eq. 41) satisfies $dG/db = F$ (checked symbolically). Quadrature of eq. (8) at $M \rightarrow 0$ reproduces σ_{KN} at $E = 1, 3, 8$ MeV to 10^{-6} . Mass suppression at $E = 3$ MeV: $\sigma/\sigma_{\text{KN}} = 0.995, 0.886, 0.609$ for $M = 0.1, 0.5, 1$ MeV.

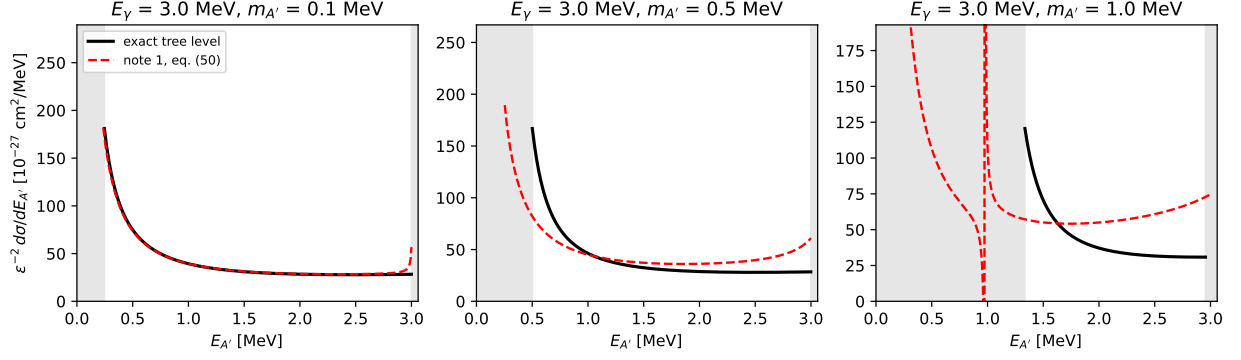


Figure 1: $\epsilon^{-2}d\sigma/dE_{A'}$ at $E_\gamma = 3$ MeV. Black: eq. (8). Red dashed: the note's eq. (50), drawn over the range the note itself claims. Grey: outside the exact range $[x_-, x_+]$. At $M = 1$ MeV the note's formula has a pole at $E_{A'} = M^2/2m = 0.978$ MeV ($u - m^2 = 0$) inside its own range.

6 The Spanish note: what is wrong and how to fix it

Dark_Photon_cross_section-1.pdf (author's name omitted) follows the right route: Feynman rules $\rightarrow$ trace $\rightarrow$ lab phase space. Its $M = 0$ total (eq. 52) is correct and equals Klein–Nishina identically (checked with sympy). Everything that depends on $M \neq 0$ needs the fixes below. The errors compound: wrong $|\mathcal{M}|^2 \times$ wrong Jacobian $\times$ wrong range.

Item	The note says	What is wrong	Fix
Kinematics, eqs. (17)–(19)	solves $M^2 + 2m(\omega - \omega') - 2\omega\omega'(1 - \cos\theta) = 0$, i.e. $ \vec{k}' \approx \omega'$; $\omega' = \frac{\omega + M^2/2m}{1 + \frac{\omega}{m}(1 - \cos\theta)}$	The algebra is right (eq. 19 does solve eq. 18) but eq. 18 itself is an approximation. At $E = 3$, $x = 2$, $M = 1$ MeV it gives $\cos\theta = 0.832$; exact is 0.960.	Keep $ \vec{k}' = \sqrt{\omega'^2 - M^2}$: replace eq. (18) by eq. (4). Work in ω' , not θ ; the relation is transcendental in ω' and does not need to be inverted.
Threshold, Table 1	$\omega \rightarrow M$	Wrong: the final state must carry the photon's momentum, which costs $M^2/2m$. Table 1 is not used in the derivation of eq. (50), so by itself this entry is harmless; the same error enters through the range, next row.	$E_{\text{th}} = M + M^2/2m$, eq. (6): 1.98 MeV for $M = 1$ MeV, not 1 MeV.
Range, eqs. (24)–(25)	$\omega' \in [\frac{m\omega + M^2/2}{2\omega + m}, \omega]$	Wrong at both ends: $\omega' = \omega$ is impossible for $M \neq 0$, and the lower end is below the true one. $E = 3$, $M = 1$: exact [1.336, 2.955], note [0.312, 3.000]. This range is non-empty for every $\omega \geq M/2$, and with $\omega' \geq M$ it gives $\omega \geq M$: it is the naive threshold of Table 1 in disguise, and this copy <i>is</i> used when eq. (50) is integrated.	Use $x_\pm$ of eq. (5) (boost of the CM momentum). For $M \rightarrow 0$ it reduces to the note's limits.

Item	The note says	What is wrong	Fix
$ \mathcal{M} ^2$, eqs. (35)–(40)	$2\epsilon^2 e^4 \left[-\frac{u}{s} - \frac{s}{u} + 2M^2 \left(\frac{1}{s} + \frac{1}{u} \right) - \frac{M^4}{4} \left(\frac{1}{s} + \frac{1}{u} \right)^2 su \right]$ ($s, u =$ propagator denominators)	Off by -39% to $+33\%$ against eq. (7). Eqs. (35)–(36) drop the m^2 terms “ $O(m_e^2/s)$ ”, but m_e^2/s is not small at MeV energies: at $M = 0$ exactly $8m^2(\frac{1}{a} + \frac{1}{b}) + 8m^4(\frac{1}{a} + \frac{1}{b})^2$ is missing, and that is what produces the $-\sin^2 \theta$ of Klein–Nishina. Eq. (39) is not the trace it names.	Replace eq. (40) by eq. (7) with $a = s, b = u$ of the note’s notation. Keep every power of m ; there is no regime in this problem where $m_e \ll \sqrt{s}$.
$d\sigma/d\Omega$, eqs. (43)–(46)	KN bracket $+M^2 f(s, u)$, $f = \frac{2}{s+u} (1 - \frac{m^2 M^2}{su})$, massless phase-space factor $(\omega'/\omega)^2$	Does not follow from eq. (40): the $-\sin^2 \theta$ term needs the m^2 terms dropped earlier. The phase-space factor for a massive final particle is $ \vec{k}' \omega'/\dots$, not ω'^2 .	Do not go through $d\Omega$. Use the invariant form $d\sigma/dt = \mathcal{M} ^2/[16\pi(s-m^2)^2]$ and $dt = 2m d\omega'$; this gives eq. (8) with no angular Jacobian at all.
Jacobian, eqs. (47)–(48)	$d\Omega/d\omega' = \frac{2\pi}{\omega'^2} (m + \frac{M^2}{2\omega})$	Derived from the approximate eq. (20); the exact one has $ \vec{k}' $ in it.	Not needed once eq. (8) is used.
Final $d\sigma/d\omega'$, eq. (50)	$\frac{\pi\epsilon^2\alpha^2}{m\omega^2} (1 + \frac{M^2}{2m\omega})[\dots]$	Off by up to $2\times$ pointwise (Fig. 1); has a pole at $\omega' = M^2/2m$ ($u = 0$ in the note’s notation) which lies inside the note’s range for $M = 1$ MeV ($0.978 \in [0.312, 3]$), so the total does not exist there. Over the exact range the total is 0.5%, 7%, 36% too high for $M = 0.1, 0.5, 1$ MeV.	Eq. (8) on $[x_-, x_+]$. The pole is outside the exact range for every $M < 2m$.

Which of these errors matters for the number of dark photons. Folding the note’s eq. (50), on the note’s range, with Park’s spectrum through eq. (9) ($\sigma_{\text{tot}} = \sigma_{\text{KN}}$, 1 GW, $\epsilon = 1$) and comparing with the same fold of eq. (8) on $[x_-, x_+]$ (`work/check_eq50_fold.py`):

M [MeV]	photons with $M < \omega < E_{\text{th}}$	$d\dot{N}/dE_{A'}$, note/exact at 3 MeV	$d\dot{N}/dE_{A'}$, note/exact at 5 MeV	$\dot{N}_{A'}(3\text{--}8 \text{ MeV})$, note/exact
0.1	1.1%	1.06	1.05	1.05
0.5	23.6%	1.48	1.43	1.46
1.0	65.9%	2.05	1.90	1.99

In the 3–8 MeV window the threshold/range error contributes nothing: an A' of $E_{A'} \geq 3$ MeV comes from a photon of $\omega \geq 3$ MeV, and all of those are above threshold for $M \leq 1$ MeV ($\omega'_{\text{max}} = \omega$ instead of x_+ moves the lower photon energy from 3 to 3.05 MeV). The factors 1.05, 1.46, 1.99 come from eq. (50) itself: the $|\mathcal{M}|^2$ without its m^2 terms and the massless Jacobian. That is the error to fix for TEXONO’s CsI. The range error matters at low $E_{A'}$: for $M = 1$ MeV, 66% of the photons above 1 MeV are below the true threshold and the note lets them produce A' ; for $E_{A'}$ between 0.31 and 1.34 MeV eq. (50) integrates over a kinematically forbidden region that contains the pole at 0.978 MeV (at $E_{A'} = 1$ MeV: exact 0, note $1.5 \times 10^{21} \text{ MeV}^{-1}\text{s}^{-1}$). That matters for a low-threshold detector such as TEXONO’s PCGe, not for the CsI.

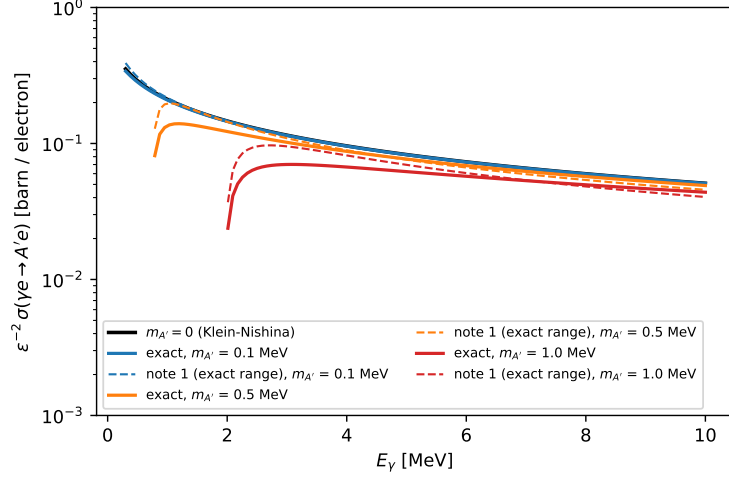


Figure 2: Total production cross section per electron, $\epsilon = 1$. Solid: eq. (8) integrated over $[x_-, x_+]$; the $M = 0$ curve is Klein–Nishina. Dashed: the note’s integrand integrated over the *exact* range (over its own range it diverges for $M = 1$ MeV).

Minimal correction of eq. (50). Replace the bracket [...] by $F/2$ with $a = 2m\omega$, $b = M^2 - 2m\omega'$, $c = m^2$, $d = M^2$; replace the prefactor $\frac{\pi\epsilon^2\alpha^2}{m\omega^2}(1 + \frac{M^2}{2m\omega})$ by $\frac{\pi\epsilon^2\alpha^2}{m\omega^2}$, i.e. $d\sigma/d\omega' = \epsilon^2 e^4 F / (32\pi m\omega^2)$; replace the range by $[x_-, x_+]$. The threshold then comes out by itself and the pole is outside the range for every $M < 2m$.

With those replacements the note’s Section 9 summary becomes: $\cos\theta$ from eq. (4), $|\overline{\mathcal{M}}|^2 = \epsilon^2 e^4 F$ with eq. (7), $d\sigma/d\omega'$ from eq. (8), range $[x_-, x_+]$ from eq. (5), threshold eq. (6). Its Section 8 (the $M = 0$ total) stays as it is.

7 The v2 report: correct, and what it does not settle

report_park_production_v2_en.pdf derives eq. (7), eq. (8), the exact range (5) and the threshold (6), with polynomial and matrix checks. All of that is confirmed here independently. It then folds the cross section with Park’s spectrum through Park’s source equation

$$\frac{d\dot{N}_{A'}}{dx} = \int dE S_\gamma(E) \frac{1}{\sigma_{\text{tot}}(E)} \frac{d\sigma(E, x)}{dx}, \quad S_\gamma = 0.58 \times 10^{18} \frac{P}{\text{MW}} e^{-E/0.91 \text{ MeV}} \text{ s}^{-1} \text{ MeV}^{-1}, \quad (9)$$

with $\sigma_{\text{tot}} = \sigma_{\text{KN}}$ as Park does. I reproduce the photon integral, $\int_{1 \text{ MeV}}^\infty S_\gamma = 1.7588 \times 10^{20} \text{ s}^{-1}$ at 1 GW (Park prints 1.76×10^{20}).

The fold itself is where *v2* and Park part company with eq. (9). Evaluating eq. (9) with the triple-checked cross section, at 1 GW, $\epsilon = 1$, $M = 0.1$ MeV (script section 9):

$E_{A'}$ [MeV]	eq. (9)	$S_\gamma(E_{A'})$	Park Fig. 1 (read by eye)	<i>v2</i> Fig. 2 (read by eye)
1	0.108	0.193	≈ 0.40	≈ 0.10
2	0.017	0.064	≈ 0.065	≈ 0.065
3	0.0036	0.021	≈ 0.022	≈ 0.020
4	0.0009	0.0072	≈ 0.007	≈ 0.005

units $10^{21} \text{ MeV}^{-1} \text{ s}^{-1}$

Above 2 MeV Park’s curve is the photon spectrum S_γ to within the reading error, and *v2*’s curve follows Park (that is its “4% median residual”). Eq. (9) cannot give that: a photon of energy E spreads its A' over $[x_-, x_+]$ with density $\sigma_{\text{KN}}^{-1} d\sigma/dx \approx 0.15\text{--}0.4 \text{ MeV}^{-1}$, so $d\dot{N}/dx \approx 0.3 \int_x^\infty S_\gamma \approx 0.27 S_\gamma(x)$ at $x = 2$ MeV, which is what the script finds (bounds $[0.009, 0.025]$ contain 0.017). The

integral of eq. (9) over x equals the number of photons to 0.1%, so the normalisation is not the issue; the shape is. Whatever Park’s generator did, it is not eq. (9); and since $v2$ matches Park rather than eq. (9), its fold is not confirmed either. Its cross section is. I did not have $v2$ ’s code to find where its fold departs.

The σ_{tot} in eq. (9) is a real difference between papers: Park uses the free-electron Klein–Nishina cross section; Smirnov et al., NEON and Lou–Wu use the XCOM total (photoelectric + Compton + pair) for the core material. Only the total makes sense as a normalisation — any interaction removes the photon. For uranium the ratio $\sigma_{\text{KN}} N_A Z/A / (\mu/\rho)_{\text{NIST}}$ tabulated in $v2$ is 0.62, 0.60, 0.43, 0.29 at 1, 3, 5, 8 MeV (recalled from $v2$ ’s table, not recomputed here). Park’s choice overestimates the Compton-like yield by 1.6–3.5 in the TEXONO window.

8 The $v3$ report and the two mechanisms

`report_darkphoton_v3_en.pdf` does not use a Compton cross section for production at all. It uses Danilov–Demidov–Gorbunov:

$$\frac{d\dot{N}_X}{dE} = P_{\gamma \rightarrow X}(E) \frac{d\dot{N}_\gamma}{dE}, \quad P_{\gamma \rightarrow X} = \frac{\epsilon^2 m_X^4}{(\Delta m^2)^2 + E^2 \Gamma^2}, \quad (10)$$

$$\Delta m^2 = \sqrt{(m_X^2 - m_\gamma^2)^2 + 4\epsilon^2 m_X^2 (m_X^2 + m_\gamma^2)},$$

with $m_\gamma^2 = 4\pi\alpha n_e/m_e \approx (20 \text{ eV})^2$ and $1/\Gamma \approx 10 \text{ cm}$. For $m_X \gg m_\gamma$ and $m_X^2 \gg E\Gamma \sim (3 \text{ eV})^2$ this is $P = \epsilon^2$ with $E_X = E_\gamma$.

This is a different mechanism, not a different derivation of the same one. In the basis where A' couples to the current with ϵe , eq. (10) in the ϵ^2 limit is direct emission of an A' at every vertex that emits a reactor photon (nuclear de-excitation, capture, annihilation): probability ϵ^2 per photon, same energy, no recoil. Lou–Wu 2026 compute exactly this for the strongest capture lines and find it beats the Compton-like channel. Park’s eq. (9) is instead a *secondary* process: a photon that already exists Compton-scatters and the outgoing photon is replaced by an A' , with the Compton energy loss. Demidov–Gorbunov–Polonski 2025 treat this same secondary channel inside the oscillation formalism (“Compton-scattered photons”) and find an enhancement factor $f(E) = 1.2$ at 2 MeV falling towards 1 above 4 MeV.

The two mechanisms give the same order of magnitude in total ($\approx \epsilon^2 N_\gamma$) but very different spectra. Computed here with Park’s spectrum, $P = 1 \text{ GW}$, $\epsilon = 1$, counting A' with $x \in [3, 8] \text{ MeV}$:

$$\begin{aligned} \dot{N}_{[3,8]}^{(A)} &= 2.53 \times 10^{18} \text{ s}^{-1} \quad (\text{Compton-like, eq. (9), } \sigma_{\text{tot}} = \sigma_{\text{KN}}), \\ \dot{N}_{[3,8]}^{(B)} &= 1.95 \times 10^{19} \text{ s}^{-1} \quad (\text{eq. (10), } P = \epsilon^2), \end{aligned} \quad (11)$$

ratio 0.130, essentially independent of M between 0.1 and 1 MeV (Figure 3). Mechanism A is not small everywhere: below 0.5 MeV it is comparable to B, because that is where the Compton-scattered A' from the many 1–2 MeV photons land. In the TEXONO window it is 13% of B. That 13% is the same size as Demidov et al.’s $f(E) - 1$ at 3–4 MeV. It is nowhere near the factor 10 claimed by Du et al. 2024; Demidov et al. trace Du’s factor to treating each scattering as a full wave-function collapse (missing the factor 1/2 and $\Gamma_{\text{abs}} \rightarrow \Gamma_{\text{tot}}$). I did not rerun either Monte Carlo; the single-scattering estimate above is consistent with Demidov and inconsistent with Du, which is as far as this check goes.

So: Park’s approach (and the $v2$ report, and NEON 2025 which copies Park’s eq. 2) computes the subleading channel and omits the leading one. That is the biggest difference among the final formulas in `apuntes/`, and it is an omission of physics, not an algebra error.

9 The detection factor 2/3

Park’s eq. (6) writes $d\sigma_{A'e \rightarrow \gamma e}/dE_r \approx \frac{2}{3}\epsilon^2 d\sigma_C/dE_r$; Lou–Wu eq. (11) repeats it; Danilov et al. call it irrelevant; the $v3$ report uses $P_{X \rightarrow \gamma} \sigma_{\text{KN}}$ with no 2/3. Computed here with the same explicit

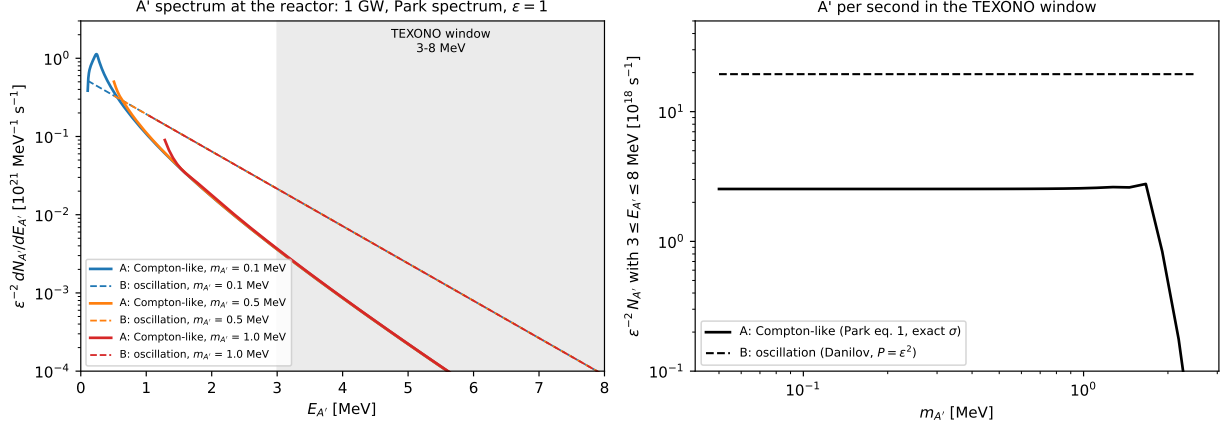


Figure 3: Expected number of A' produced per second at a 1 GW reactor with Park's spectrum, $\epsilon = 1$. Left: spectrum $d\dot{N}_{A'}/dE_{A'}$ for mechanism A (Compton-like, eq. (9) with the exact cross section, solid) and mechanism B (oscillation, eq. (10) with $P = \epsilon^2$ and $E_{A'} = E_\gamma$, dashed) for $m_{A'} = 0.1, 0.5, 1$ MeV; grey band: TEXONO's 3–8 MeV window. Right: A' per second in that window versus mass. B exceeds A by 7.7 in the window; both are $\propto \epsilon^2$.

matrices for the inverse process $A'(q) + e \rightarrow \gamma + e$, integrated over the CM angle, $E_{A'} = 3$ MeV, $M = 0.01$ MeV:

$$\frac{\sigma_T}{\epsilon^2 \sigma_{\text{KN}}} = 0.99999, \quad \frac{\sigma_L}{\epsilon^2 \sigma_{\text{KN}}} = 3 \times 10^{-5}, \quad \frac{\frac{1}{3}(2\sigma_T + \sigma_L)}{\epsilon^2 \sigma_{\text{KN}}} = 0.66667. \quad (12)$$

The $2/3$ is the unpolarized average: two transverse polarizations that scatter like photons and one longitudinal that does not. It applies only to an unpolarized A' beam. The reactor beam is not unpolarized: oscillation/direct emission gives transverse A' only (Danilov: longitudinal production is $\propto m_X^2/E^2$), and Compton-like production, computed here from the explicit longitudinal vector, gives a longitudinal fraction $\sigma_L/\sigma_{\text{tot}} = 0.010$ at $(E, M) = (3, 0.1)$ MeV, 0.195 at $(3, 0.5)$, 0.078 at $(3, 1)$, 0.211 at $(8, 1)$ (Fig. 4). The correct detection cross section is

$$\sigma_{\text{det}} = f_T \sigma_T + f_L \sigma_L, \quad \frac{\sigma_T}{\epsilon^2 \sigma_{\text{KN}}} = 0.98\text{--}1.00, \quad \frac{\sigma_L}{\epsilon^2 \sigma_{\text{KN}}} = 0.02\text{--}0.31 \quad (M = 0.5\text{--}1, E_{A'} = 3\text{--}8 \text{ MeV}), \quad (13)$$

with $f_{T,L}$ the produced fractions. For $M \lesssim 0.1$ MeV this is $\epsilon^2 \sigma_{\text{KN}}$ to 1%: factor 1, not $2/3$. Danilov is right, Park is wrong, and Lou–Wu's eq. (11) is a correct statement about an unpolarized beam applied to a beam that is not. In ϵ the $2/3$ is a 10% effect ($\epsilon \propto N^{-1/4}$); it matters for the bookkeeping, not for the exclusion plot.

10 The formula to use

For the number of A' per unit energy arriving at a detector at distance R , per second, for $m_{A'} \gg m_\gamma$ and away from the resonance $m_{A'} \approx m_\gamma$:

$$\left[\frac{d\Phi_{A'}}{dx} = \frac{1}{4\pi R^2} \left[\epsilon^2 \frac{m_{A'}^4}{m_{A'}^4 + x^2 \Gamma_{\text{tot}}^2} S_\gamma(x) f(x) + \int dE S_\gamma(E) \frac{1}{\sigma_{\text{tot}}(E)} \frac{d\sigma(E, x)}{dx} \right] \right] \quad (14)$$

where the first term is eq. (10) (dominant; $f(x)$ is Demidov's secondary-photon factor if the second term is not kept explicitly — do not count both), $d\sigma/dx$ is eq. (8) with eq. (7), and σ_{tot} is the XCOM total for the core material. The detector rate is then $N_e T \int dx (d\Phi/dx) \sigma_{\text{det}}(x)$ with σ_{det} as in the previous section. The $v3$ report implements the first term only, with $\sigma_{\text{det}} = \epsilon^2 \sigma_{\text{KN}}$; that

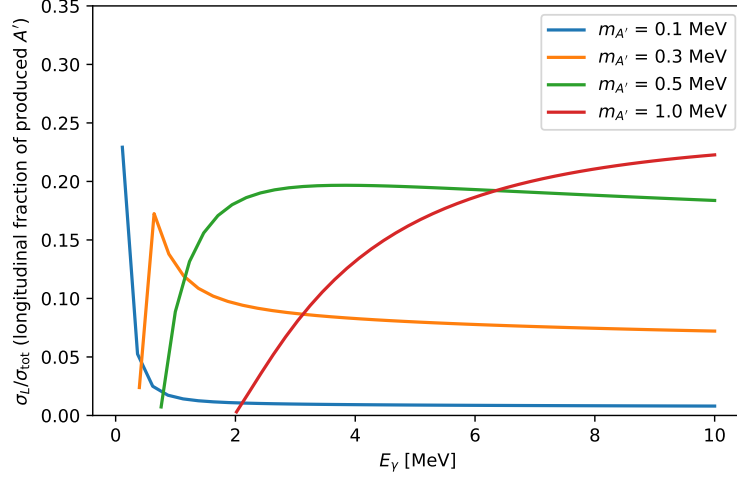


Figure 4: Longitudinal fraction of Compton-produced A' , from the explicit ξ_L contraction integrated over $[x_-, x_+]$. Not $\propto M^2/E_\gamma^2$; near threshold and for $M \gtrsim 0.3$ MeV it is 10–25%.

is the right leading order. The $v2$ report and Park implement the second term only; the Spanish note implements the second term with the wrong cross section.

Two normalisation choices in **apuntes/** are not errors but must be stated: the TEXONO event cap is 197.2 (Park, 1.96σ two-sided) or 195.7 (Danilov, one-sided 95%, used by $v3$), and Danilov multiply it by the anti-Compton suppression factor 6 to 1174.1 (also used by Du et al.); $v3$ does not.

11 What was checked and what was not

Checked, with assertions: the explicit-matrix $|\mathcal{M}|^2$ against $v2$, deNiverville and Smirnov; the Ward identity; the KN limit; the exact range and threshold; the note’s eqs. (18)–(20), (40), (50), (52) and the fold of (50) with the spectrum; the $v2$ primitive $G(b)$; the KN total by quadrature; the polarization-resolved inverse cross sections; the window counts and spectra of the two mechanisms, including that the integral of eq. (9) over $E_{A'}$ returns the photon number; Park’s photon integral.

Not checked: the Gondolo–Raffelt total (paper not in the folder; $v2$ ’s own check stands at 7×10^{-14}); Park’s generator and $v2$ ’s fold code (neither available; the Figure 1 and Figure 2 values above are read by eye from the rasters, $\pm 30\%$); the XCOM/NIST attenuation numbers (recalled from $v2$); Du’s and Demidov’s Monte Carlos; the resonance region $m_{A'} \approx m_\gamma$; bound-electron and reactor-transport effects; anything in deNiverville beyond eqs. (18)–(20) and in NEON beyond its eq. (2).

Run receipt: 21 September 2026, `work/.venv/bin/python work/check_cross_section.py` (numpy 2.5.2, sympy 1.12, scipy 1.18.1, Python 3 system, venv with marimo 0.24.2); output in `work/check_cross_section.log`, last line ALL ASSERTIONS PASSED; numbers in `work/check_cross_section_out.json`; figures by `marimo export html work/figures_cross_section.py`.

References

- [1] H. K. Park, Phys. Rev. Lett. 119, 081801 (2017). `papers_reactor_dark_photon/01`.
- [2] M. Danilov, S. Demidov, D. Gorbunov, Phys. Rev. Lett. 122, 041801 (2019). 02.
- [3] P. deNiverville, H.-S. Lee, Y.-M. Lee, arXiv:2011.03276, eqs. (18)–(20). 03.

- [4] M. Smirnov et al., Phys. Rev. D 104, 116024 (2021), eqs. (3)–(4). 04.
- [5] M. Du, J. Liu, X.-P. Wang, T. Wu, Phys. Rev. D 109, 055041 (2024). 05.
- [6] NEON Collaboration, Phys. Rev. Lett. 134, 021802 (2025), eq. (2). 06.
- [7] S. Demidov, D. Gorbunov, A. Polonski, arXiv:2512.17652. 07.
- [8] Y. Lou, L. Wu, arXiv:2606.08487, eqs. (8), (11), App. A. 08.